

Making a 512KB expansion RAM card for 6809/6800 and FLEX: FM77AV40.

 flexonsbd.blogspot.com/2022/10/fm772040512kbram.html

Friday, October 21, 2022

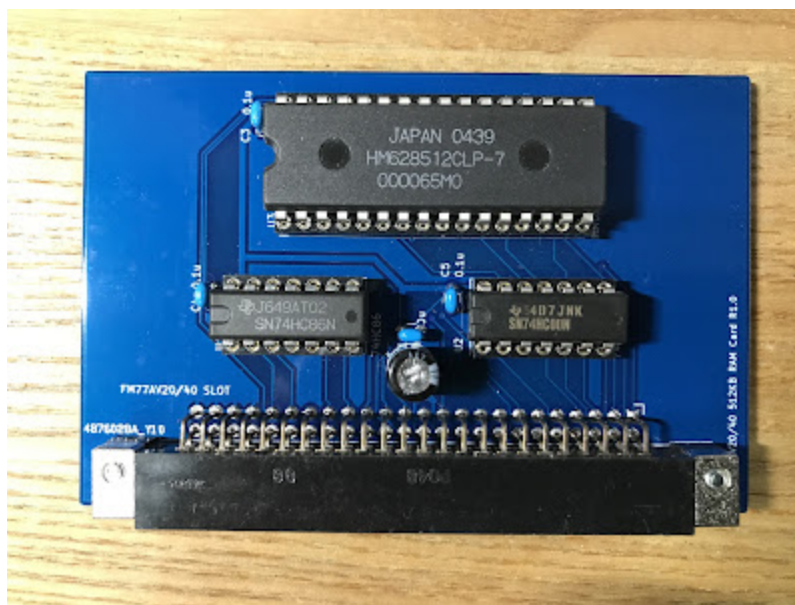
Making a 512KB expansion RAM card for the FM77AV40

I have created a 512KB expansion RAM card for the FM77AV ~~20/40~~.

[Correction October 23, 2022] We received feedback that the FM77AV20/EX only has a 40P connector, so we have corrected the compatible models to AV40/EX/SX. We have also changed the title. Thank you for pointing this out.

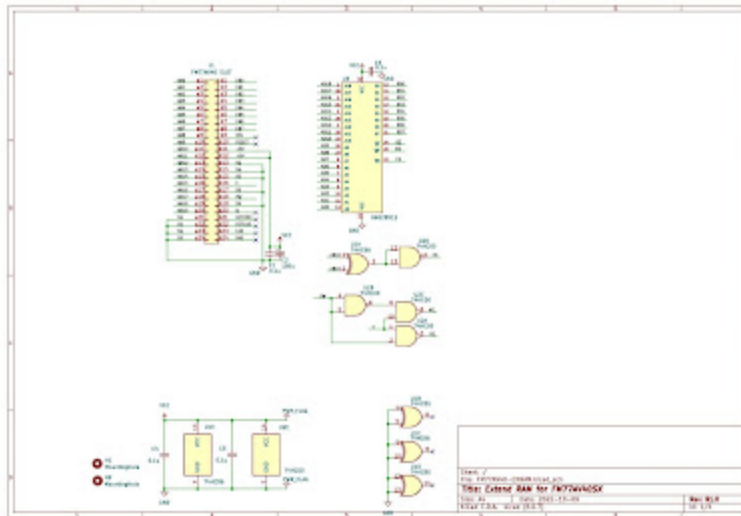
I was finally able to get my hands on a Fujitsu 48-pin connector, which had been hard to find, so I decided to make a 512KB expansion RAM card for the FM77 ~~20/40~~, something I'd wanted to do for a while.

Here is the SRAM card I made.



The 512KB SRAM card I made

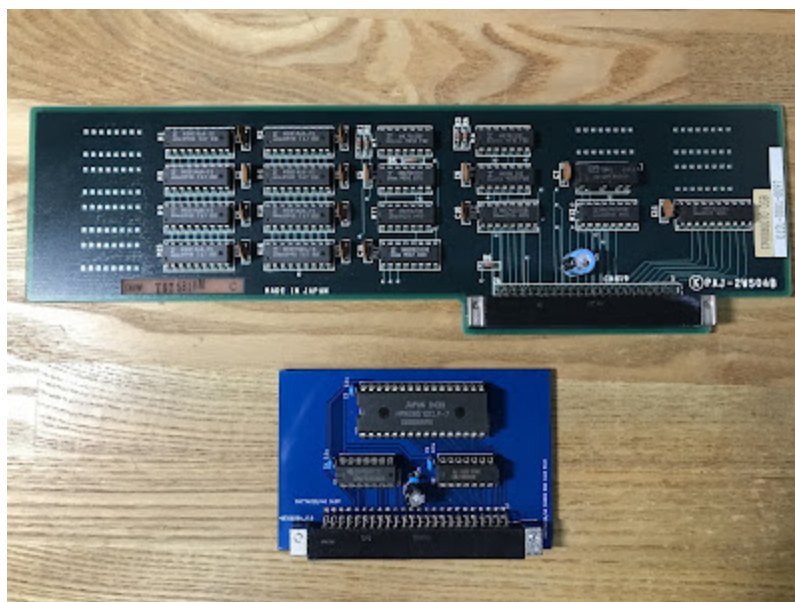
The circuit diagram is shown below.



Circuit diagram

Actually, there's one area that's not so great. I've connected the RW signal to the SRAM's WE terminal, but I meant to connect the 74HC00's WE output signal instead; I made a mistake. However, since I'm inputting the E signal to the OE signal, I don't think there will be any problems with its operation, so I've left it as is.

Besides 512KB of SRAM, only two TTL-ICs were needed, resulting in a compact size of 90x60mm.



I compared it to the original RAM card.

I haven't done a detailed operational check yet, but for now I've tried checking it with the mfree command in OS-9 Level 2.

Indeed, 512KB of capacity is reserved.



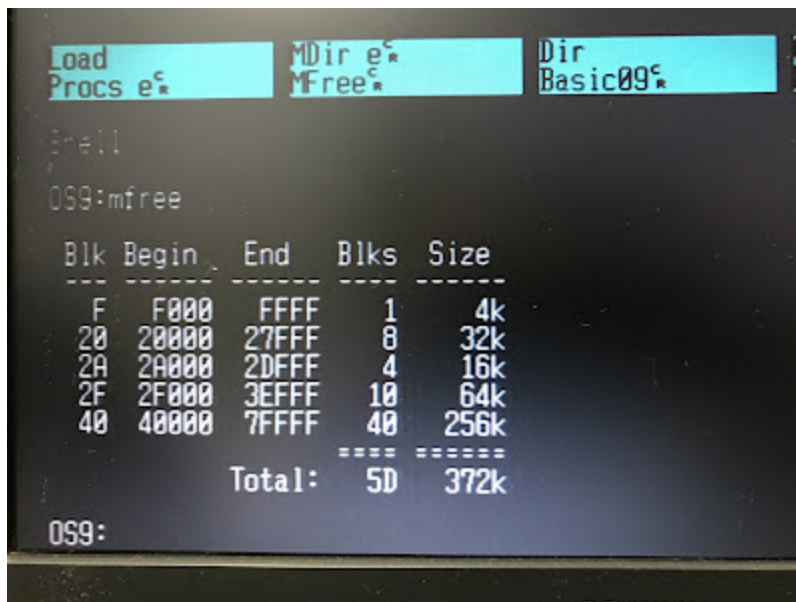
A screenshot of the OS-9 Level 2 command prompt. At the top, there are four status bars: 'Load Procs e%' (cyan), 'MDir e%' (cyan), 'Dir Basic09%' (cyan), and 'Color. Key.' (cyan). Below these, the text 'Shell' and 'OS9:mfree' are visible. A table displays memory allocation details. The table has five columns: 'Blk', 'Begin', 'End', 'Blks', and 'Size'. The data rows are as follows:

Blk	Begin	End	Blks	Size
F	F000	FFFF	1	4k
20	20000	27FFF	8	32k
2A	2A000	2DFFF	4	16k
2F	2F000	3EFFF	10	64k
40	40000	BFFFF	80	512k
Total:			9D	628k

Below the table, the prompt 'OS9:' is visible. The 'CENTURY' logo is at the bottom right of the screen.

In the case of a manufactured RAM card

The original RAM card has a capacity of 256KB.



A screenshot of the OS-9 Level 2 command prompt, similar to the one above. The status bars at the top are the same. The text 'Shell' and 'OS9:mfree' are visible. A table displays memory allocation details. The table has five columns: 'Blk', 'Begin', 'End', 'Blks', and 'Size'. The data rows are as follows:

Blk	Begin	End	Blks	Size
F	F000	FFFF	1	4k
20	20000	27FFF	8	32k
2A	2A000	2DFFF	4	16k
2F	2F000	3EFFF	10	64k
40	40000	7FFFF	40	256k
Total:			5D	372k

Below the table, the prompt 'OS9:' is visible. The 'CENTURY' logo is at the bottom right of the screen.

In the case of a genuine RAM card

I haven't completed a detailed memory test yet, but it seems to be working for now.

The point was that if you can get your hands on 512KB of SRAM and a 48-pin connector, you can easily expand your memory.